HDOC Day-36

String Operations - Part 1

Question: Write a menu-driven C program using separate functions for the operations shown in the assignment (options 1-3 and 4 to exit).

C Program:

```
#include <stdio.h>

int countCharacters(char s[]) {
    int i = 0;
    while (s[i] != '\0') i++;
    return i;
}

void printCharacters(char s[]) {
    int i;
    for (i = 0; s[i] != '\0'; i++)
        printf("%c\n", s[i]);
}

void countVowelsConsonants(char s[]) {
    int i, v = 0, c = 0;
    char ch;
    for (i = 0; s[i] != '\0'; i++) {
        ch = s[i];
        if (ch >= 'A' && ch <= 'Z') ch += 32;
        if (ch >= 'a' && ch <= 'z') {
            if (ch == 'a' || ch == 'e' || ch == 'i' ||
                ch == 'o' || ch == 'u') v++;
            else c++;
        }
    }
    printf("Vowels = %d, Consonants = %d\n", v, c);
}

int main() {
    char s[200];
    int choice;
    printf("Enter a string: ");
    fgets(s, sizeof(s), stdin);
    for (int i = 0; s[i] != '\0'; i++) {
        if (s[i] == '\n') { s[i] = '\0'; break; }
    }
    do {
        printf("\n1. Count characters");
        printf("\n2. Print each character on new line");
        printf("\n3. Count vowels and consonants");
        printf("\n4. Exit\nEnter choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1: printf("Length = %d\n", countCharacters(s)); break;
            case 2: printCharacters(s); break;
            case 3: countVowelsConsonants(s); break;
            case 4: printf("Exit\n"); break;
            default: printf("Invalid choice\n");
        }
    } while (choice != 4);
    return 0;
}
```

Sample Output: For input Hello: length = 5; characters = H, e, l, l, o (each on a new line); vowels = 2 and consonants = 3.